

# JOSHUA OLUNLADE

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## EDUCATION

### MPhil in Engineering Education

Expected 2027

University of Cape Town, Centre for Research in Engineering Education (CREE)

Supervised by A/Prof. Bruce Kloot and Dr Lauren Jansen.

- Research: Navigating Access to Industrial Placements, a constructivist grounded theory study of Nigerian engineering undergraduate students.

### M.Eng in Mechanical Engineering

Adeleke University, Ede

- Thesis: Development of a Screen-Free Robotics Kit for Enhancing Computational Thinking in Early Education in Underserved Communities. Supervised by Engr. Prof. Olutosin O. Ilori and Engr. Kehinde M. Adeleke.

### B.Eng in Mechanical Engineering

July 2022

Adeleke University, Ede

First Class (Honours)

### Senior Secondary Certificate of Education (SSCE)

2014 – 2017

Ebunoluwa Intentional School, Ofatedo

Club: Junior Engineering and Technicians Club

### Basic Education Certificate Examination (BECE)

2011 – 2014

Deeper Life High School, Osogbo

### Primary School Leaving Certificate

2011

Good Tidings Nursery and Primary School, Osogbo

## PUBLICATIONS

### Journal Articles

1. Olunlade, J., Adeleke, K., & Ilori, O. (2026). Affordable, Screen-Free Robotics for Enhancing Computational Thinking in Early STEM Education: A Comprehensive Review. *Uniosun Journal of Engineering and Environmental Sciences*, 8, 44–56. <https://doi.org/10.64980/ujees.v8i1.005>

### Conference Papers

1. Olunlade, J. O., & Olayemi, M. (2026, June). Exploring FreeCAD for Teaching Engineering Graphics (GET 102) in Developing Nations. Paper presented at *2026 ASEE Annual Conference & Exposition*, Charlotte, North Carolina. <https://doi.org/10.18260/1-2--59749>

## TEACHING EXPERIENCE

### Tutor

2026 – Present

Department of Mechanical Engineering, University of Cape Town

Tutoring, grading and related teaching duties.

- MEC1020W, Introduction to Mechanical Engineering (2026)
- MEC2050S, Thermofluids 1 (2026)

### Graduate Assistant

Jan 2025 – Jan 2026

Department of Mechanical Engineering, Adeleke University, Ede

- Supported the development of GET 102, Engineering Graphics and Solid Modeling I. Taught tutorial classes covering 100+ students and created tutorial resources to aid their knowledge of solid modelling and engineering graphics.
- Supported various classroom sessions for various courses.
- Supported departmental administration as the secretary of the departmental board.

**Robotics Instructor**

Sep 2020 – July 2022

*Jilbotics Limited*

- Facilitated online classes for a basic robotics course with a focus on Arduino.
- Designed instructional materials for both synchronous and asynchronous learning to support student understanding.
- Graded and reviewed practical assignments carried out by students.

**Peer Tutor (Informal)**

Sep 2018 – Dec 2020

*Adeleke University, Ede*

- Volunteered to provide tutorial support to peers in General Mathematics, Trigonometry and Algebra, Fluid Mechanics, and Thermodynamics.
- Assisted classmates in understanding difficult concepts and preparing for exams.
- Created YouTube Videos to enhance peer understanding.
- Organized study groups and developed learning materials to enhance group learning.

**PROFESSIONAL EXPERIENCE**

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**Equipment and Maintenance Officer**

Nov 2023 – Dec 2024

*Sphero Grid Nigeria Limited*

- Managed and maintained all equipment, including surveying and geoinformatics instruments, computer systems, and mechanical systems such as vehicles and generators.
- Conducted regular inspections, basic repairs and troubleshooting.
- Coordinated with technicians and vendors for repairs and procurement of new tools.

**Automobile Repair Training Exchange Internship**

Jul 2024 – Aug 2024

*Sphero Grid Nigeria Limited*

- Completed an eight-week training internship focusing on automobile repair and maintenance.
- Gained hands-on experience with various repair techniques and tools.
- Worked closely with experienced local mechanics to learn about vehicle systems, troubleshooting, and repair processes.
- Applied practical skills to assist in routine maintenance and repair tasks, enhancing overall understanding of automotive repair.

**Junior Production Engineer**

Nov 2022 – Oct 2023

*Federal University of Agriculture, Abeokuta (FUNAAB) Integrated Ventures*

- Contributed to the production of various goods including sachet and bottled water, garri, fufu flour, and cashew nuts within the university's internal production unit.
- Assisted in overseeing production processes to ensure efficiency and product quality.
- Supported equipment maintenance and troubleshooting to minimize downtime.
- Collaborated with the production team to improve workflow and output.

**Student Intern**

Mar 2021 – Aug 2021

*Prototype Engineering Development Institute, Ilesha*

- Developed Windows-based desktop applications using Java and the JavaFX framework.
- Designed and prototyped a small Arduino-controlled robot car, enhancing practical skills in robotics and embedded systems.

**Student Intern**

May 2020 – Oct 2020

*Department of Mechanical Engineering, Adeleke University*

- Contributed to the design and development of an innovative automatic dispensing unit with touchless stations for water, soap, and sanitizer, utilizing proximity sensing technology.
- Assisted in creating an automatic disinfectant chamber that sprays disinfectant upon entry, supporting the university's COVID-19 response.
- Gained in-depth knowledge of automatic control principles, microcontroller programming, and sensor selection.

## Student Intern

Jun 2019 – Sep 2019

*Prototype Engineering Development Institute, Ilesha*

- Developed skills in solid modeling of components using CAD software, specifically Autodesk Inventor.
- Acquired programming expertise in C and gained practical experience in hardware programming with Arduino.
- Enhanced understanding of G Code and participated in simulations related to CNC Lathe G Code operations.

## LEADERSHIP AND ENTREPRENEURSHIP

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### Founder & Executive Director

Apr 2023 – Present

*Care for Knowledge (CFK)*

- Established a nonprofit organization dedicated to providing educational resources and mentorship to students.
- Developed and implemented programs to make educational materials accessible to all, regardless of location or income.
- Led efforts in fundraising, partnership building, and community outreach to support the organization's mission and expand its impact.

### Project Lead

Jan 2023 – Present

*Morship Game Development*

- Conceptualized and launched a Christian board game designed to foster fellowship through interactive gameplay.
- Led project development, including design, production, and marketing efforts.
- Managed a team, coordinated resources, and ensured project alignment with educational and entertainment goals.

### Founder & Director

Jan 2023 – Present

*Sphero Life Limited*

- Co-founded Sphero Life Limited, focusing on creating educational and interactive content for Christian communities.
- Directed project planning, content development, and community outreach.
- Oversaw strategic initiatives, budget management, and organizational growth.

## HONOURS, AWARDS AND FUNDING

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### Intra-Africa Mobility Grant

*Capacity Building for Engineering Education (CB4EE) consortium*

Funding for MPhil studies.

### Best Graduating Student

2022

*Department of Mechanical Engineering, Adeleke University, Ede*

### Microsoft Office Specialist National Championships, Word 2019 Category

2022

## PROFESSIONAL MEMBERSHIPS

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### Graduate Member (GNSE)

2022 – Present

*Nigerian Society of Engineers*

## CERTIFICATIONS AND TRAINING

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### Digital Manufacturing & Design

July 2024

*Coursera (Offered by University at Buffalo, The State University of New York)*

- Explored the shift from paper-based to digital processes in manufacturing. Gained insights into how digital manufacturing and design (DM&D) transforms product design and production, enhances company responsiveness, and creates new career opportunities. Learned about the impact of technology on manufacturing practices and how it increases competitiveness and productivity.

[Certificate link](#)

### How to Create an Online Course

Nov 2023

*Coursera (Offered by The University of Edinburgh)*

- Understood how to design learner centered instructional materials.
- Completed practical activities on producing educational content for learners.

[Certificate link](#)

### How to Create Videos for Online Course

Nov 2023

*Coursera (Offered by The University of Edinburgh)*

- Developed relevant skills to script, storyboard, and film educational videos.
- Developed an understanding of the various tools and resources needed for the pre-production, production, and post-production phases in the creation of online courses.

[Certificate link](#)

### Viral Marketing and How to Craft Contagious Content

Nov 2023

*Coursera (Offered by The University of Pennsylvania)*

- Developed an understanding of the principles that make ideas sticky and viral.
- Understood the concepts of how social influences shape human behavior and the power of word of mouth.
- Understood the power of social networks in making content spread.

### Microsoft Excel Expert (Microsoft 365 Apps and Office 2019)

July 2022

*Issued by Microsoft*

[Certificate link](#)

### Microsoft Word Expert (Microsoft 365 Apps and Office 2019)

July 2022

*Issued by Microsoft*

[Certificate link](#)

### Microsoft Office Specialist: PowerPoint Associate (Office 2019)

April 2022

*Issued by Microsoft*

[Certificate link](#)

## COMMUNITY SERVICE

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### Discipleship Coordinator

Nov 2023 – Feb 2026

*Grace Bible Mission Students' and Teenagers' Church*

- Coordinated discipleship programs for students and teenagers, fostering spiritual growth and community involvement.
- Organized and led discipleship meetings, Bible study sessions, and mentorship activities.
- Developed and implemented strategies to engage youth and enhance their spiritual development.
- Collaborated with other church leaders to support and guide the younger members of the congregation.

**Team Leader & Technical Support**

Feb 2024 – Jan 2025

*WikiQuestions*

- Led a team in the collaborative sharing of past examination papers to aid students' exam preparation.
- Provided technical support for the platform, ensuring smooth operation and resolving any issues.
- Coordinated with team members to enhance platform functionality and user experience.
- Developed and implemented strategies to increase user engagement and expand the platform's reach.

**Team Member**

Nov 2022 – Oct 2023

*Gender Equity and Charity, Community Development Service Group (NYSC)*

- Participated in outreach programs to educate the community on gender equity and charitable practices.
- Assisted in organizing events and workshops focused on raising awareness and promoting gender equality.
- Contributed to developing educational materials and engaging with community members to foster understanding and support.

**National General Secretary**

Sep 2020 – Sep 2022

*Grace Bible Mission Students' Fellowship*

- Coordinated the overall administrative functions of the fellowship, ensuring smooth operations and effective communication.
- Managed and generated detailed reports on fellowship activities and progress.
- Assisted with organizing events and meetings, facilitating a cohesive and well-organized fellowship environment.

**Student Chaplain**

Sep 2019 – May 2022

*Wisdom Chapel, Adeleke University, Ede*

- Supported the chapel's growth and development by assisting the Pastor with member records and spiritual activities.
- Engaged in various tasks assigned by the Pastor to contribute to the chapel's mission and community outreach.
- Facilitated member engagement and supported spiritual development initiatives.

**National Bible Study Secretary**

Sep 2019 – Sep 2021

*Grace Bible Mission Students' Fellowship*

- Prepared and organized Bible study materials for both offline and online sessions.
- Coordinated and facilitated Bible study meetings, contributing to spiritual growth and engagement within the fellowship.
- Performed other duties associated with the role to support the fellowship's mission and activities.

**Organizing Team Member**

2019

*TEDxOsogbo, 2019*

- Contributed to the planning and execution of TEDxOsogbo, a high-profile conference featuring speakers from various fields.
- Assisted with logistics, including venue setup, scheduling, and speaker coordination.
- Collaborated with the organizing team to ensure a smooth event experience for attendees and speakers.
- Supported marketing and outreach efforts to increase event visibility and audience engagement.

**REFERENCES**

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Available on request